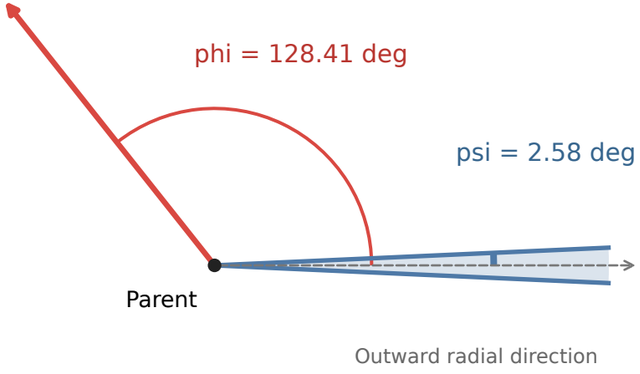
Why the child direction is still outside the cone

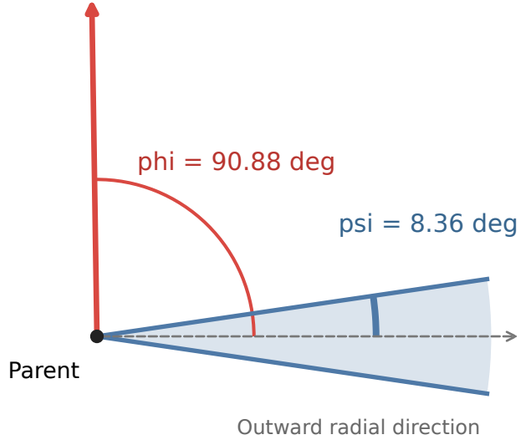
Local-angle sketches at the parent point, using measured mean angles

Original C0
lambda=.1, c=1, tau=.07



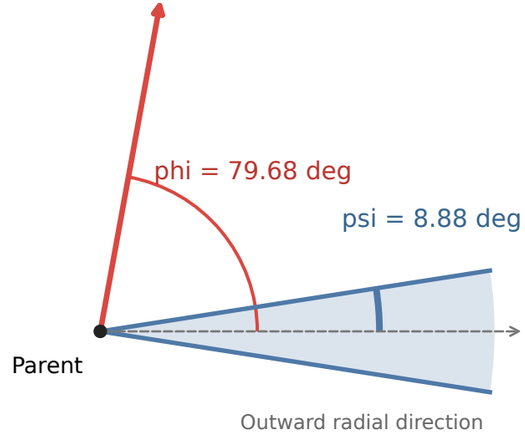
Mean excess = 125.84 deg

Current fixed
lambda=.2, c=1, tau=.035



Mean excess = 82.52 deg

Current learned
lambda=.2, c=.6171, tau=.04365



Mean excess = 70.80 deg

Blue wedge: cone directions | Red arrow: direction toward the child at the parent | Inclusion requires phi <= psi

Schematic only: arrow lengths are arbitrary, and averaged angles do not represent an individual sample or a 2D embedding projection.